RETAIL ASSORTMENT OPTIMIZATION

Data-driven insights for revenue growth and smarter product placement

Prepared by: Natalya Rtishcheva

Business Challenge

Assortment Optimization Strategy

- Segment products into Core and Complementary categories.
- Leverage cross-selling mechanics and checkout zone optimization.
- Anchor high-velocity impulse items against stable, planned hardware purchases.

Data Overview

Dataset Specifications.

- Initial volume: 6,737 transaction records.
- Core attributes: date, customer_id, order_id, product, quantity, price.
- External enrichment: automated web scraping and category mapping.

Data Preparation

Advanced Preprocessing Pipeline

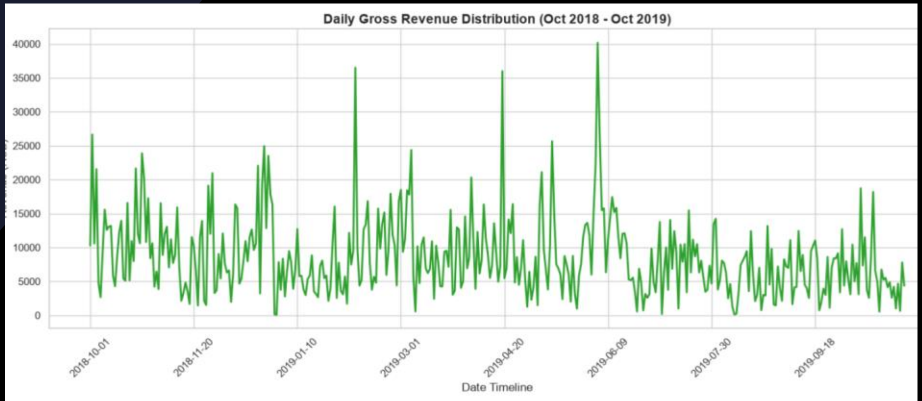
- Standardized multi-format headers and data types.
- Executed Two-Factor 99th Percentile Filtering for Quantity and Price.
- Removed 2.02% of wholesale anomalies (e.g., single orders of 1,000 items).
- Final clean retail dataset: 6,485 rows.

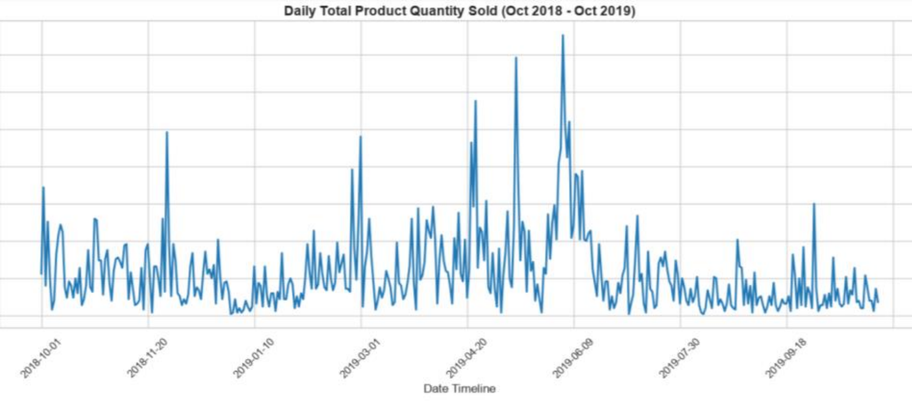
```
QUANTITY THRESHOLD: 26.6 units
PRICE THRESHOLD: 4,225.4 c.u.

ANOMALIES DROPPED: 136 rows (2.02%)
FINAL DATASET: 6,485 rows
```

Exploratory Analysis

Key Purchase Dynamics

- Sales Volume: Massive traffic peak in May–June driven by Gardening category.
- Sales Revenue: High-value spikes in Oct/Dec/Feb driven by premium holiday decor.
- Basket Profiles: Distinct right-skewed distribution.
- Retail Baseline: Median Check is 698.00 c.u.; Average Check is 1,265.07 c.u..

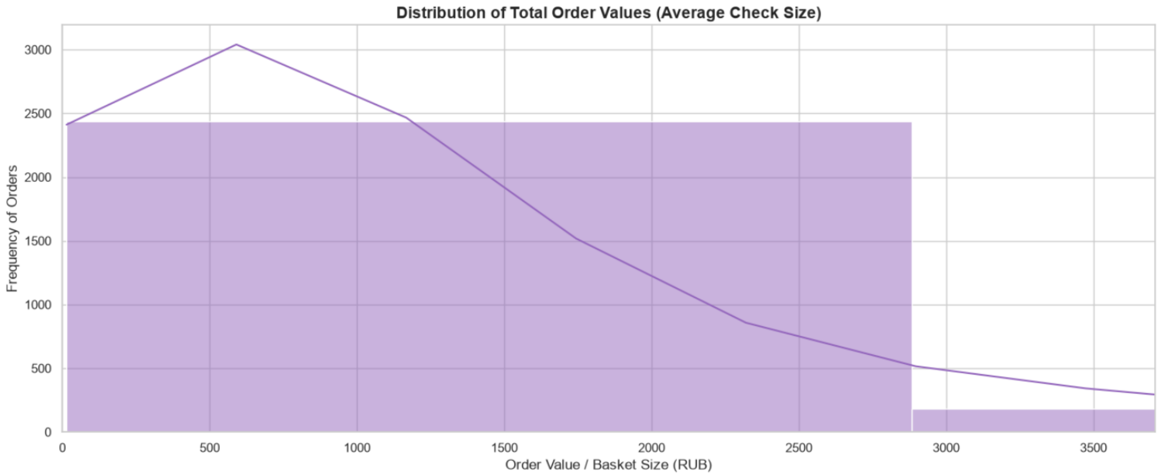




Statistical Insights

Hypothesis Testing & Assortment Split

- D'Agostino-Pearson test rejected price normality ($p = 0.0$), shifting model to non-parametric tests.
- Mann-Whitney U test confirmed highly significant variance between groups ($p = 0.0$).
- Core Segment (Home, Plumbing): High-ticket, planned purchases (Median: 750.00 c.u.).
- Complementary Segment (Gardening, Floristry, Other): Rapid impulse volume (Median: 120.00 c.u.).



Market Basket Analysis & Recommendation Engine

- Built a binary transaction matrix utilizing optimized Pearson correlation coefficients.
- Model Audit Target: "Rassada Kabachka sort Zebra" (Absolute top-selling product).
- Top Co-Purchase Discoveries: Zucchini "Belogor" (0.61), "Sosnovsky" (0.57), "Zolotinka" (0.55).
- Metric confirms strong variety-seeking behavior within the agricultural category.

Machine Learning Extension

Business Recommendations

Actionable Merchandising Strategy

- Place high-velocity Complementary items (Median: 120 c.u.) at checkout registers for impulse buys.
- Implement Group-Display Merchandising: group correlating seedling variations together on shelves.
- Auto-suggest related accessories online when high-ticket Core equipment (3,000+ c.u.) is selected.

Business Impact

Expected Commercial Value

- Shelf Layout Optimization: Up to 3x increase in secondary items per targeted basket.
- Strategic Bundles: Projected 3–5% seasonal revenue uplift via crop multi-packs.
- Inventory Planning: Reduced stock-outs during May-June volume peaks.
- Basket Size Expansion: Gradual organic growth of the low 698 c.u. median baseline.

Tools & Skills

Technical Stack

- Python: pandas, numpy (data engineering), matplotlib, seaborn (visual analytics).
- Web Scraping: requests, BeautifulSoup (external assortment taxonomy enrichment).
- Advanced Analytics: SciPy stats (Mann-Whitney U test, D'Agostino normality diagnostics).
- Data Science Model: Optimized Pearson transaction mapping engine.

Thank You for Your Attention!

I will be glad to answer your questions.

- Presented by: Natalya Rtishcheva
- Email: natartispro@gmail.com
- GitHub: https://github.com/Rnvart